

Can Egocentric Pose Reconstruction Support Trustworthy Home Rehabilitation? A Cross-Dataset Study of Kinematic Reliability and Feedback Risk

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Abstract. Wearer-pose reconstruction from head-mounted cameras could make home rehabilitation assessment unobtrusive, but low average joint error does not establish that the resulting exercise scores or corrective feedback are dependable. We present EgoRehab-Bench, a cross-dataset protocol that connects egocentric 3D pose error to the reliability and safety of downstream rehabilitation decisions. The protocol first measures reconstruction accuracy on genuine egocentric video with motion-capture ground truth. It then replays the observed, temporally correlated residuals over ground-truth movements from REHAB24-6 and UCOPhyRehab, preserving joint, visibility, and motion-speed dependencies. This separation permits rehabilitation-specific evaluation without mischaracterizing external-camera datasets as egocentric. We evaluate geometric accuracy, joint-angle agreement, repetition segmentation, exercise-quality classification, and structured feedback. Our primary safety outcome is the rate of corrections that are issued from reconstructed motion but unsupported by ground-truth motion. Reliability is quantified using intraclass correlation, standard error of measurement, minimal detectable change, Bland–Altman agreement, participant-level confidence intervals, and calibrated abstention. Across $_$ reconstruction models and $_$ datasets, the best configuration obtains an MPJPE of $_$ mm and an unsafe false-feedback rate of $_$ %, while retaining $_$ % assessment coverage. [The final abstract will state which exercises and measurements met the registered fitness-for-purpose criteria.](#) The study provides a reproducible route from pose-estimation benchmarks to claims that are meaningful for home rehabilitation, and identifies conditions in which an automated coach should withhold feedback rather than present an uncertain correction.

Keywords: Egocentric vision · human pose estimation · home rehabilitation · reliability · movement assessment · feedback generation

1 Introduction

Home rehabilitation shifts repeated practice from a supervised clinic to an environment in which direct observation is intermittent or absent. A camera-based system could measure motion, identify exercise errors, and return immediate guidance without requiring markers or a full motion-capture laboratory. Conventional third-person video, however, depends on a camera being placed far enough away to see the whole body. That requirement is inconvenient in small rooms, sensitive to furniture and camera orientation, and easy to violate between sessions. A camera worn on the head moves with the user and may therefore offer a more portable sensing configuration.

Egocentric sensing also changes the estimation problem. A downward-facing fisheye camera observes the wearer’s body with severe perspective distortion, frequent self-occlusion, and large changes in visibility. An outward-facing head-mounted camera may not observe the body at all and must infer motion from head trajectory, scene structure, or learned motion priors. Recent systems reconstruct increasingly plausible full-body motion from these limited observations [15,18,3,2,19,7]. Sparse-tracking methods similarly infer the body from the head and hands and now model field-of-view loss and global-position shifts explicitly [9,8]. These advances motivate rehabilitation applications, but they do not yet answer the decision that matters in practice: is a reconstruction reliable enough to judge a particular exercise and issue a particular correction?

The usual answer is based on Mean Per-Joint Position Error (MPJPE). MPJPE is necessary for comparing pose estimators, but it averages errors across joints, time, subjects, and movements. Two reconstructions with the same MPJPE can have different clinical consequences. A nearly constant translation may have little effect on a knee-flexion angle, whereas a smaller but directionally structured ankle error may reverse a decision about knee-over-toe alignment. A temporally smooth bias may preserve repetition counting but distort range of motion. Brief tracking failures can be almost invisible in a frame-averaged metric while causing a false correction at the precise instant a movement reaches its endpoint. Reliability therefore cannot be inferred from aggregate geometric accuracy alone.

This distinction is consistent with evidence from clinical markerless motion capture. Reviews find that validity and reliability vary by joint, plane, task, and system, and that strong performance for spatiotemporal measures does not imply interchangeable kinematics [11,13]. A home coaching system adds a further transformation: estimated pose is converted into exercise features, those features into findings, and findings into language. Each transformation can amplify, attenuate, or hide upstream error. An evaluation that stops at joint positions cannot quantify that risk.

No single public dataset resolves the problem. Egocentric pose datasets provide paired first-person observations and 3D ground truth, but their motion inventories are not organized around physiotherapy and generally lack expert exercise-quality labels. REHAB24-6 and UCOPhyRehab provide rehabilitation movements and reference joint data [5,1], but their RGB views are external. Treating those videos

as first-person inputs would invalidate the central experimental claim. A sound evaluation must preserve the separate evidential role of each dataset.

We introduce EgoRehab-Bench, a benchmark and analysis protocol that connects these sources without conflating them. First, we run reconstruction methods on genuine egocentric video and retain their complete residual trajectories rather than only reporting an average. Second, we align and replay these residual trajectories over ground-truth rehabilitation movements. The replay preserves temporal correlation and conditions on joint visibility and motion speed. Third, we compare rehabilitation measurements, decisions, and feedback produced from reconstructed motion with those produced from the reference motion. This design estimates the downstream sensitivity to error processes observed in real egocentric reconstruction. It does not claim to replace a future dataset that records rehabilitation simultaneously with head-mounted video and laboratory motion capture.

Our evaluation is organized around a fitness-for-purpose criterion. A method is not declared suitable merely because it ranks first on MPJPE. It must show acceptable agreement for the target kinematic measurement, a minimal detectable change below a registered task-specific tolerance, a low rate of unsupported corrections, calibrated confidence, and useful coverage after uncertain cases are withheld. This formulation also permits a negative result: some exercises may support repetition counting but not joint-specific feedback, and some body regions may remain unsuitable for autonomous assessment.

Taken together, this paper makes four contributions:

- (1) We formulate egocentric rehabilitation assessment as a chain from video to reconstructed pose, clinical features, structured findings, and feedback, and define reliability at each interface.
- (2) We introduce a cross-dataset error-replay protocol that combines genuine egocentric reconstruction benchmarks with REHAB24-6 and UCOPhyRehab while preserving the acquisition semantics of every dataset.
- (3) We provide rehabilitation-oriented metrics that complement MPJPE, including joint-angle agreement, ICC, SEM, MDC95, decision agreement, unsafe false-feedback rate, and coverage-risk curves under calibrated abstention.
- (4) We release a reproducible evaluation implementation and a registered interpretation rule that determines when a configuration may support autonomous feedback and when it must abstain.

After the experiments, the final paragraph will summarize the central quantitative finding and identify the exercises for which the reliability criteria were and were not met.

2 Related Work

2.1 Egocentric Reconstruction of the Camera Wearer

Egocentric body reconstruction spans several sensor configurations. xR-EgoPose estimates 3D pose from a headset-mounted camera and introduced a large

synthetic dataset for the task [15]. Mo2Cap2 demonstrated real-time inference from a cap-mounted fisheye camera, using the characteristic top-down view of the wearer’s body [18]. Later work improved robustness in unconstrained scenes and global coordinates [16]. UnrealEgo expanded the scale and visual diversity of synthetic egocentric data [3], while UnrealEgo2 and UnrealEgo-RW added stereo perception, hand annotations, and a larger real-world test setting [2]. These datasets are the appropriate source for direct claims about image-based egocentric reconstruction.

A related line reconstructs motion from sparse signals supplied by current headsets. AvatarPoser predicts full-body pose from head and hand observations [9]. EgoPoser improves robustness to global displacement, intermittent hand visibility, temporal ambiguity, and variation in body shape [8]. More recent systems combine head-mounted images, visual localization, scene geometry, and generative motion priors [19,7]. These methods demonstrate that plausible motion can be recovered when the body is only partly visible or not visible. Plausibility, however, can conceal clinically consequential errors: a generated squat can look natural while assigning an incorrect knee angle to the wearer. We therefore treat image-based and sparse-tracking methods as distinct input regimes and do not compare them unless their required observations are available under the same protocol.

Existing egocentric benchmarks emphasize MPJPE, aligned MPJPE, velocity error, and qualitative appearance. The metrics are appropriate for reconstruction, but they do not state whether derived measurements are stable across repeated estimates or whether a downstream decision changes. Our study retains the standard metrics for comparability and adds an application-level evaluation.

2.2 Vision-Based Rehabilitation Assessment

Markerless pose estimation has been investigated as an accessible input for exercise recognition, kinematic measurement, telerehabilitation, and coaching. The attraction is straightforward: commodity cameras reduce instrumentation and can capture motion in a patient’s usual environment. The evidence base is nevertheless heterogeneous, and clinical measurement remains an early application area [11]. Variation in cameras, populations, tasks, reference systems, and statistical reporting makes broad statements about accuracy difficult.

REHAB24-6 was designed to examine pose estimation for six rehabilitation exercises. It provides synchronized external RGB video, 2D and 3D joints, repetition intervals, and correct/incorrect executions [5]. Its cross-subject, cross-camera, and cross-exercise settings make it useful for testing whether an exercise-analysis pipeline generalizes beyond familiar participants and viewpoints. UCOPhyRehab combines five external RGB cameras with OptiTrack reference measurements for rehabilitation movements [1]. The multi-view design supports analysis of camera sensitivity, while the later UCOPhyRehab++ annotations add per-repetition ordinal assessments. Neither dataset contains wearer-mounted full-body video. In this work they therefore supply rehabilitation motion and decision labels, not direct egocentric images.

REST-HANDS is a notable exception: it records hand rehabilitation using smart glasses and evaluates exercise recognition, form, and repetition counting [12]. Its clinical motivation and first-person acquisition are closely aligned with our question, but its hand-focused label space does not test full-body reconstruction. We include it as an optional external task rather than use it to generalize conclusions to lower-limb exercise.

Recent work has also explored language models for exercise assessment and feedback on pose sequences [14]. Language generation can improve the clarity and flexibility of feedback, but it introduces another potential source of error. Our pipeline separates biomechanical inference from realization in language. A structured finding must first be supported by the motion signal; a language model, when used, may paraphrase that finding but may not introduce a new diagnosis or correction.

2.3 Accuracy, Agreement, and Reliability

Accuracy, agreement, and reliability answer different questions. A pose estimator can have low mean error but inconsistent repeated measurements, or high correlation with a reference despite a systematic bias. Bland and Altman showed why correlation is not evidence that two measurement methods agree [4]. Reliability studies consequently report both relative indices such as intraclass correlation and absolute indices such as the standard error of measurement and minimal detectable change [17,10].

Markerless motion-capture reviews reinforce the need for outcome-specific analysis. Reported validity varies substantially across joints and movement planes, and distal or occlusion-prone joints often remain difficult [13]. For rehabilitation, the relevant threshold is not a universal number of millimetres. It is the amount of error that changes an interpretation, exceeds an accepted measurement tolerance, or causes misleading feedback. EgoRehab-Bench operationalizes this distinction by carrying uncertainty through the complete decision chain.

2.4 Confidence and Selective Prediction

A home system cannot assume that every frame is assessable. Clothing, lighting, self-occlusion, limited field of view, and rapid motion can make an estimate under-determined. A safer policy is selective prediction: issue a result only when estimated risk is below a threshold and otherwise abstain. Neural-network confidence is not automatically calibrated, so calibration must be measured rather than assumed [6]. We report both error among accepted assessments and the coverage retained at each threshold. This exposes whether an apparently safe method achieves low risk only by refusing nearly all difficult cases.

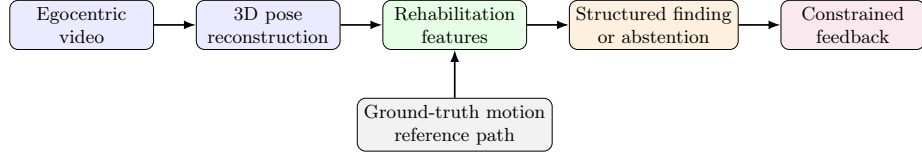


Fig. 1. Evaluation chain. Reconstruction quality is measured on genuine egocentric data. Empirical residual trajectories are then replayed on rehabilitation motion, and the resulting features and feedback are compared with a ground-truth reference path.

3 Problem Formulation

We consider a head-mounted camera system that records a monocular or synchronized stereo video sequence

$$\mathcal{I}_{1:T} = \{I_t^{(c)} \mid t = 1, \dots, T; c = 1, \dots, C\}, \quad (1)$$

where $C \in \{1, 2\}$. A reconstruction model g_θ maps a temporal window of images and optional camera calibration to a body pose

$$\hat{\mathbf{P}}_t = g_\theta(\mathcal{I}_{t-w+1:t}, \mathbf{K}, \mathbf{D}), \quad \hat{\mathbf{P}}_t \in \mathbb{R}^{J \times 3}, \quad (2)$$

where $\mathbf{K}$ and $\mathbf{D}$ denote intrinsic and fisheye-distortion parameters. Depending on the benchmark, the output is represented in a camera, device, pelvis, or global coordinate system. Every result is converted to a declared analysis frame before comparison; alignment is never applied silently.

Rehabilitation assessment is represented as two further mappings. A feature extractor ϕ converts the pose sequence into interpretable measurements such as repetition boundaries, joint range of motion, symmetry, or trunk compensation,

$$\hat{\mathbf{z}} = \phi(\hat{\mathbf{P}}_{1:T}), \quad (3)$$

and a registered decision function h converts measurements and uncertainty into a structured finding $\hat{\mathbf{y}}$,

$$\hat{\mathbf{y}} = h(\hat{\mathbf{z}}, \hat{\sigma}; \tau). \quad (4)$$

Here τ contains exercise-specific thresholds fixed on training or validation data. The decision function can return an abstention symbol $\perp$ when uncertainty exceeds the registered limit. Natural-language realization r is applied only after Eq. 4; it cannot add findings not present in the structured record.

Figure 1 summarizes the evaluation. The reference path computes $\mathbf{z}^* = \phi(\mathbf{P}^*)$ and $\mathbf{y}^* = h(\mathbf{z}^*, \mathbf{0}; \tau)$ from ground-truth motion. The test path computes the same outputs from reconstructed or error-replayed motion. Comparing both paths isolates the downstream consequences of pose error.

Table 1. Dataset roles. “Direct” means that a reconstruction model is evaluated from its intended egocentric image input. “Replay” means that measured egocentric residual trajectories are applied to reference rehabilitation motion. Counts will be verified against the downloaded releases before submission.

Dataset	View	Reference	Labels	Role
UnrealEgo2	stereo ego	3D body/hand	motion	direct
UnrealEgo-RW	stereo ego	3D pose	motion	direct
REHAB24-6	two external RGB	2D/3D joints	rep., binary	replay
UCOPhyRehab	five external RGB	OptiTrack points	exercise	replay
UCOPhyRehab++	multi-view	3D joints/angles	ordinal scores	replay
REST-HANDS	glasses RGB	task labels	form, count	optional direct

3.1 Research Questions

The study addresses four research questions.

RQ1: Reconstruction validity. How accurately and consistently do head-mounted video methods reconstruct rehabilitation-relevant joints under changes in visibility, motion speed, body region, and real versus synthetic capture?

RQ2: Measurement reliability. How much do empirically observed egocentric errors change joint angles, range of motion, temporal events, and symmetry measures on REHAB24-6 and UCOPhyRehab?

RQ3: Decision and feedback risk. How often do reconstruction errors change exercise-quality labels or produce an unsupported corrective message, and which uncertainty thresholds control that risk?

RQ4: Fitness for purpose. Which combinations of model, exercise, outcome, and capture condition satisfy all registered criteria for autonomous assessment, and which require abstention or clinician review?

We expect distal, frequently occluded joints to produce larger absolute error and a higher feedback-risk contribution than visible proximal joints. This directional hypothesis is frozen before test-set evaluation. The study does not depend on it being confirmed.

4 Datasets and Experimental Roles

Table 1 distinguishes datasets by acquisition geometry, reference signal, and role in the study. This distinction is central to the validity of the protocol. Direct reconstruction accuracy is reported only where the model receives the first-person observation for which it was designed. Rehabilitation datasets with external cameras are used to test measurement and decision sensitivity.

4.1 UnrealEgo2 and UnrealEgo-RW

UnrealEgo2 contains synchronized stereo fisheye observations from a virtual glasses rig and provides camera-relative body and hand annotations. It extends UnrealEgo with additional motions and temporal sequences [3,2]. UnrealEgo-RW

provides paired real-world stereo observations and motion-capture reference data. We use the official released test splits and the camera calibration distributed by the authors. Sequence identity, human model identity, and environment identity are retained in every output row so that uncertainty can be aggregated at an appropriate level.

Synthetic and real data are never pooled without a domain indicator. Synthetic data offer scale and exact annotations, whereas real data include optical artifacts, clothing, imperfect segmentation, and calibration error. We report both domains separately and use real-world residuals as the primary source for the conservative rehabilitation replay. Synthetic residuals support a higher-powered secondary analysis of motion and visibility strata.

The initial motion inventory is screened without inspecting model error. Motions are mapped to broad rehabilitation-relevant primitives: shoulder elevation, arm abduction, elbow flexion, hip abduction, squat-like flexion, lunge-like stance, forward reach, walking, and transitions between standing and floor-supported poses. The mapping is recorded before quantitative evaluation. Results are shown for the complete official test split and for the pre-declared rehabilitation-relevant subset.

4.2 REHAB24-6

REHAB24-6 contains ten participants performing six rehabilitation exercises under physiotherapist instruction, with correct and deliberately incorrect repetitions [5]. Two synchronized external RGB cameras observe the subject from different directions, while motion capture supplies 3D joints. The release also contains temporal segmentation and exercise-error descriptions. These properties make it the primary dataset for binary correctness and repetition-level feedback analysis. Our audit of the official segmentation file found 1,072 annotated repetitions from 65 paired recordings: 568 labelled correct and 504 labelled incorrect. Fifteen segments carry the release’s erroneous-motion-capture flag and are excluded by the primary quality-control rule rather than repaired after outcome inspection.

We preserve the official subject and camera identifiers. All views of one participant remain in the same cross-validation fold. The two camera recordings do not constitute independent samples. When an exercise can be performed on either side, the body is canonicalized by active side only after the label has been read from metadata. No mirror augmentation is applied to test data.

REHAB24-6 does not contain head-mounted images. Accordingly, we do not report an “egocentric MPJPE on REHAB24-6.” Instead, its ground-truth joint trajectories are used as the clean reference path in Fig. 1. Empirical residual windows from the egocentric experiments are transformed into the REHAB24-6 body frame and replayed over these trajectories. This tests whether error with the observed magnitude, direction, intermittency, and temporal structure alters rehabilitation decisions.

4.3 UCOPhyRehab and UCOPhyRehab++

UCOPhyRehab was recorded from 27 adults in a controlled indoor environment using five RGB cameras and an OptiTrack reference system [1]. Markers were placed on the shoulder, elbow, and wrist for upper-limb exercises or on the hip, knee, and ankle for lower-limb exercises. We therefore evaluate only measurements supported by the reference points available for each exercise. Missing full-body joints are not imputed and do not enter MPJPE denominators.

The original dataset is used for continuous joint and angle agreement across camera configurations. The downloaded UCOPhyRehab++ metadata contains 411 subject-exercise records and 1,992 scored repetitions from 27 subjects across 16 exercise identifiers. The released scores are strongly concentrated at 4 and 5; only two repetitions have a score of 2 and none has a score of 1. We therefore report class support explicitly and do not collapse the ordinal task into an apparently balanced accuracy problem. The primary agreement statistic is quadratic weighted kappa; rank correlation is reported only as association. **[TODO: Confirm with the dataset authors whether the released scalar score is an individual rating, an adjudicated rating, or an aggregation of multiple physiotherapists.]**

As with REHAB24-6, the RGB cameras are external. Their views help characterize how a conventional system behaves, but they are not inputs to the egocentric reconstructor in the primary study. A secondary comparison may run a third-person pose estimator on these images to quantify the trade-off between an easily placed wearable camera and a carefully placed room camera. That comparison is labelled by sensing configuration and is not interpreted as an egocentric ablation.

4.4 Optional Direct Rehabilitation Benchmark

REST-HANDS records first-person hand exercises using smart glasses [12]. If access is granted, it provides a direct test of exercise recognition, form classification, and repetition counting from egocentric video. It does not provide the full-body 3D reference required by RQ1 and is analyzed as a separate hand-rehabilitation case study. Conclusions from REST-HANDS are not generalized to lower-limb kinematics.

4.5 Data Governance

Only data permitted by the respective research licenses are processed. Raw videos are kept in access-controlled cluster storage and are not redistributed. Derived error summaries are released only when their source license permits it and when they cannot be inverted to reconstruct identifiable imagery. Dataset versions, checksums, access dates, and license texts are recorded in the experiment manifest.

5 Cross-Dataset Reliability Protocol

5.1 Coordinate and Skeleton Harmonization

The datasets and reconstruction methods use different joint sets, coordinate frames, units, and alignment conventions. We define a common evaluation skeleton containing the pelvis, neck, bilateral shoulders, elbows, wrists, hips, knees, ankles, and feet. Only joints supported by both a prediction and its reference enter a comparison. Each adapter declares its source joint, any midpoint construction, axis permutation, sign change, unit conversion, and parent relation in a versioned configuration file.

Let $\mathbf{P}_t^d \in \mathbb{R}^{J_d \times 3}$ be a pose in dataset d . A deterministic adapter A_d maps it into the common skeleton and preserves a binary availability mask $\mathbf{m}_t^d$:

$$(\mathbf{P}_t, \mathbf{m}_t) = A_d(\mathbf{P}_t^d). \quad (5)$$

The mask prevents absent joints from being interpreted as zero-valued observations. Bone lengths, left-right orientation, and a neutral-pose fixture are tested for every adapter. We retain predictions in the native coordinate frame and generate pelvis-aligned and Procrustes-aligned copies only for metrics that explicitly require them.

Joint angles are computed from vectors meeting at a joint. For points $\mathbf{a}, \mathbf{b}, \mathbf{c}$, the unsigned angle at $\mathbf{b}$ is

$$\alpha(\mathbf{a}, \mathbf{b}, \mathbf{c}) = \cos^{-1} \left(\frac{(\mathbf{a} - \mathbf{b})^\top (\mathbf{c} - \mathbf{b})}{\|\mathbf{a} - \mathbf{b}\|_2 \|\mathbf{c} - \mathbf{b}\|_2 + \epsilon} \right). \quad (6)$$

Signed plane-specific angles use a body-centred basis derived from the pelvis and shoulders. Frames with a degenerate basis are flagged rather than silently repaired.

5.2 Direct Reconstruction Evaluation

For each genuine egocentric test sequence, model outputs are evaluated using the official checkpoint and preprocessing. We store frame-level residuals

$$\mathbf{E}_{t,j} = \hat{\mathbf{P}}_{t,j} - \mathbf{P}_{t,j}^* \quad (7)$$

before aggregation. Alongside each residual, we store model confidence, ground-truth visibility, predicted visibility, motion speed, acceleration, sequence identity, subject or avatar identity, and domain. This record is the interface between the direct reconstruction and rehabilitation analyses.

The minimum model comparison contains the public UnrealEgo baseline and the temporal stereo method released with UnrealEgo2 [3,2]. A monocular baseline is added only if its required calibration and output frame can be reproduced from the released data. We do not compare against a sparse head-and-hand method using synthetic inputs derived from ground truth, because doing so would give that method observations unavailable from the raw-video system.

5.3 Temporally Correlated Error Replay

Frame-independent Gaussian perturbation would underestimate the impact of sustained bias, smooth drift, and tracking loss. We instead sample contiguous residual windows from Eq. 7. For a rehabilitation sequence with body-frame rotation $\mathbf{R}_t$ and reference pose $\mathbf{Q}_t^*$, a sampled residual window $\mathbf{E}_{s:s+L-1}$ is applied as

$$\tilde{\mathbf{Q}}_{t,j} = \mathbf{Q}_{t,j}^* + \gamma \mathbf{R}_t \mathbf{E}_{s+t,j}, \quad t = 0, \dots, L-1, \quad (8)$$

where $\gamma = 1$ in the primary analysis. The residual source is matched by joint, visibility, and motion-speed stratum. A sensitivity sweep varies γ to determine the error magnitude at which each downstream outcome becomes unreliable.

Residuals are centered only in analyses explicitly labelled “zero-bias.” The primary analysis preserves systematic bias because it is part of real measurement error. Tracking gaps are reproduced using the original missingness mask. Short gaps are interpolated only if the same rule is used by the deployed reconstruction pipeline; long gaps trigger missing features or abstention.

After perturbation, a single constrained forward-kinematics projection restores the reference subject’s bone lengths while minimizing displacement from $\tilde{\mathbf{Q}}$. This step removes anatomically impossible limb stretching without erasing joint-angle error. Results without this projection are reported as an ablation.

For each rehabilitation sequence, $B = 100$ residual windows are drawn with a fixed random seed. Sampling is hierarchical: a source motion identity is drawn first, then a valid contiguous window. This prevents long sequences from dominating the error distribution. Every replay retains the source residual identifier for audit.

5.4 Exercise Feature Extraction

The feature set is defined before inspecting test labels. Each feature has a joint dependency, unit, temporal phase, direction, and validity rule. The core features are:

- shoulder elevation and abduction range;
- elbow flexion range and terminal extension;
- hip abduction and frontal-plane trunk compensation;
- knee flexion at the deepest squat or lunge phase;
- bilateral knee and hip symmetry;
- trunk inclination and lateral lean;
- repetition duration, eccentric/concentric duration, and inter-repetition variability.

Repetitions are segmented by a finite-state machine operating on the primary exercise angle and its angular velocity. Entry and exit thresholds are selected on validation subjects, with minimum-duration and hysteresis constraints fixed across the test set. The automatic boundaries are compared with dataset annotations using tolerance-window F1 and absolute count error.

Feature uncertainty is estimated by an ensemble, predictive distribution, or temporal augmentation according to the reconstruction model. When a method provides no native uncertainty, we calibrate a post-hoc regressor on validation residuals using confidence, visibility, motion speed, and temporal inconsistency. The test set is used only once after the calibration rule and threshold have been frozen.

5.5 Structured Findings and Feedback

Each exercise has a registered rubric that maps features to structured findings. A finding contains the exercise, movement phase, anatomical variable, direction, magnitude, confidence, and evidence frames. Thresholds are estimated from correct training executions or supplied by the dataset’s expert rubric. Thresholds are never tuned on test correctness labels.

The primary feedback realization is deterministic. For example, a bilateral knee flexion deficit during the terminal squat phase maps to a single concise instruction that names the observable movement and an actionable adjustment. The system suppresses duplicate messages within a repetition and prioritizes at most one finding according to registered clinical importance and confidence.

An optional language-model condition receives only the structured finding and an approved phrase inventory. Its output is rejected if it changes the body side, exercise phase, direction, or recommended action. This design tests whether linguistic realization can improve readability without allowing the language model to reinterpret noisy coordinates.

5.6 Selective Assessment

Let u_i be the calibrated probability that assessment i is wrong. The system accepts the assessment when $u_i \leq \tau_u$ and otherwise returns $\perp$. For a set of N assessments, coverage and selective risk are

$$\text{Coverage}(\tau_u) = \frac{1}{N} \sum_{i=1}^N \mathbb{K}[u_i \leq \tau_u], \quad (9)$$

$$\text{Risk}(\tau_u) = \frac{\sum_i \ell(\hat{y}_i, y_i^*) \mathbb{K}[u_i \leq \tau_u]}{\sum_i \mathbb{K}[u_i \leq \tau_u] + \epsilon}, \quad (10)$$

where ℓ is zero-one loss for categorical findings and a registered tolerance loss for continuous measures. We report the complete coverage-risk curve and mark the pre-registered operating point.

6 Evaluation

6.1 Reconstruction Metrics

We report MPJPE in the native declared frame, pelvis-aligned MPJPE, and PA-MPJPE after rigid Procrustes alignment. These metrics are not interchangeable:

pelvis alignment removes global translation, while Procrustes alignment also removes rotation and scale. We additionally report MPJVE and mean acceleration error to quantify temporal quality, per-joint error, and the proportion of frames for which a required joint is missing or below confidence.

For rehabilitation-relevant angles, we report mean absolute error, signed bias, the 95th percentile absolute error, and endpoint error at the exercise phase used for a decision. Aggregate results are macro-averaged over motion identities and then over subjects. A frame-weighted value is included only as a secondary diagnostic.

6.2 Agreement and Reliability

For a continuous feature z , agreement with the reference is summarized by the mean difference and 95% limits of agreement

$$\bar{d} \pm 1.96s_d, \quad d_i = \hat{z}_i - z_i^*, \quad (11)$$

with participant-level bootstrap confidence intervals [4]. We test proportional bias by regressing the difference on the pairwise mean. If proportional bias is detected after false-discovery-rate correction, regression-based limits are reported instead of a single constant band.

Repeated error replays permit estimation of ICC(2,1), a two-way random-effects, absolute-agreement coefficient [10]. We also report

$$\text{SEM} = s_{\text{pooled}}\sqrt{1 - \text{ICC}}, \quad \text{MDC}_{95} = 1.96\sqrt{2}\text{SEM}, \quad (12)$$

in raw units and as a percentage of the group mean [17]. ICC is never reported without its 95% confidence interval, SEM, and MDC95. Reliability bands are interpreted from the lower confidence bound, not only from the point estimate.

6.3 Exercise Decisions

Binary correctness is evaluated using balanced accuracy, precision, recall, specificity, F1, and Matthews correlation coefficient. We report participant-macro and exercise-macro estimates with 95% bootstrap intervals. For ordinal UCO-PhyRehab++ scores, we report the confusion matrix, quadratic weighted kappa, and mean absolute score error. Classification thresholds are frozen using nested, subject-exclusive validation.

Repetition segmentation is evaluated by onset and offset error, boundary F1 within ± 0.5 s, count mean absolute error, and off-by-one accuracy. A repetition is the sampling unit for feedback metrics; frames within a repetition are not treated as independent observations.

6.4 Feedback Outcomes

Let F_i^* be the set of findings obtained from reference motion and $\hat{F}_i$ the findings obtained from reconstructed motion for repetition i . We define the unsafe false-feedback rate as

$$\text{UFFR} = \frac{\sum_i |\hat{F}_i \setminus F_i^*|}{\sum_i |\hat{F}_i| + \epsilon}. \quad (13)$$

The complementary missed-feedback rate is computed from $F_i^* \setminus \hat{F}_i$. Exact-match accuracy is intentionally not the only measure because a repetition may contain more than one valid finding. We therefore report set precision, set recall, micro and macro F1, and agreement on body side, movement phase, direction, and priority.

Calibration is evaluated using Brier score, expected calibration error with bins fixed before test evaluation, and reliability diagrams [6]. The primary system-level plot is risk against coverage as the abstention threshold varies. The registered operating point must satisfy $\text{UFFR} \leq \text{---}\%$ while retaining coverage $\geq \text{---}\%$.

6.5 Statistical Model

For continuous error outcomes, we fit a mixed-effects model with fixed effects for model, exercise, body region, visibility, motion-speed quartile, and domain, and random intercepts for participant or motion identity and source residual sequence. Interactions between model and body region and between model and visibility are pre-specified. Skewed non-negative errors are log transformed after adding a declared numerical constant, or analyzed with a Gamma mixed model if residual diagnostics reject the Gaussian specification.

For binary feedback errors, we use a mixed-effects logistic model with the same fixed effects and random intercept structure. Effect sizes are reported as marginal probabilities and risk differences with confidence intervals. The paper emphasizes estimation rather than null-hypothesis significance. The limited family of confirmatory contrasts is corrected using Holm’s method; all other contrasts are labelled exploratory.

6.6 Splits and Leakage Controls

All preprocessing choices, exercise thresholds, uncertainty calibrators, and abstention thresholds are selected without test-subject labels. Views of the same person and repetitions from the same recording remain in one fold. Residual windows used to tune the replay or confidence model come from training and validation identities only. The primary cross-dataset result uses real egocentric residuals and rehabilitation subjects unseen during any threshold selection.

We report model-selection performance separately from final test performance. If a public checkpoint was trained on any evaluated sequence, that overlap is documented and the affected result is excluded from the primary generalization claim.

Table 2. Direct reconstruction performance on genuine egocentric video. Values are participant- or motion-macro estimates with 95% confidence intervals. Lower is better.

Dataset	Method	MPJPE (mm)	PA-MPJPE (mm)	MPJVE (mm/s)	Failure (%)
UnrealEgo2	UnrealEgo baseline	—	—	—	—
UnrealEgo2	Stereo temporal	—	—	—	—
UnrealEgo-RW	UnrealEgo baseline	—	—	—	—
UnrealEgo-RW	Stereo temporal	—	—	—	—

6.7 Fitness-for-Purpose Decision

A model-exercise-outcome configuration is considered suitable for autonomous feedback only if all of the following hold:

1. the upper 95% confidence bound of absolute measurement error is below the registered task-specific tolerance;
2. MDC95 is smaller than the published clinical anchor or the protocol-defined meaningful difference;
3. the upper 95% confidence bound of UFFR is below the registered safety limit; and
4. coverage at that safety limit is above the registered usability minimum.

If no defensible clinical anchor exists, the outcome is labelled “unresolved” rather than suitable. This prevents a statistically precise but clinically uninterpretable measurement from being presented as validated.

7 Results

RESULTS PENDING. This section defines the tables and comparisons that will be populated automatically after the registered analysis. No value in this section is an estimate from real data.

7.1 Egocentric Reconstruction Accuracy

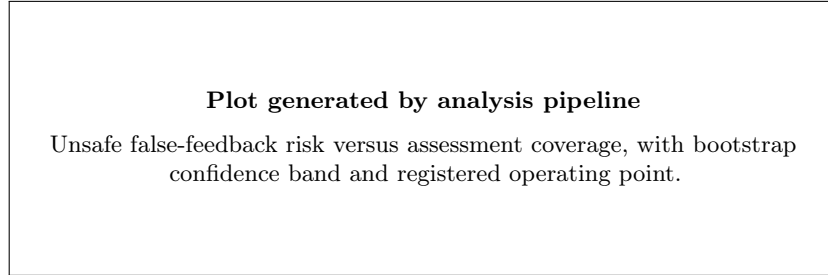
Across — test sequences from — motion identities, the evaluated models processed — frames. Table 2 reports the direct reconstruction results. The primary model obtained an MPJPE of — mm on UnrealEgo2 and — mm on UnrealEgo-RW. Error was largest at the [joint/body region determined by the analysis](#), and increased from — mm to — mm between the lowest and highest motion-speed quartiles.

7.2 Rehabilitation Measurement Reliability

Table 3 summarizes agreement after empirical error replay. On REHAB24-6, — of — registered features met the measurement-error tolerance. On UCOPhyRehab, — of — supported joint-angle measurements met the criterion. The median MDC95 was —° for proximal angles and —° for distal angles. [State whether error differed by visibility and whether the directional hypothesis was supported.](#)

Table 3. Illustrative structure of the reliability table. Every point estimate will be accompanied by a 95% confidence interval. LoA denotes limits of agreement.

Dataset	Outcome	MAE	Bias	LoA	ICC(2,1)	MDC95
REHAB24-6	shoulder ROM	—	—	—	—	—
REHAB24-6	knee flexion	—	—	—	—	—
UCOPhyRehab	elbow ROM	—	—	—	—	—
UCOPhyRehab	hip abduction	—	—	—	—	—

**Fig. 2.** Coverage-risk curve for feedback under calibrated abstention. The final figure will show each reconstruction method and dataset separately.

7.3 Exercise Decisions and Feedback

Using ground-truth skeletons, the registered exercise classifier achieved a balanced accuracy of $\underline{\quad}\%$ on REHAB24-6. After egocentric error replay, balanced accuracy was $\underline{\quad}\%$, a difference of $\underline{\quad}$ percentage points. Repetition-boundary F1 changed from $\underline{\quad}$ to $\underline{\quad}$. These two values distinguish limitations of the exercise model from errors introduced by pose reconstruction.

Without abstention, UFFR was $\underline{\quad}\%$ and missed-feedback rate was $\underline{\quad}\%$. At the pre-registered uncertainty threshold, UFFR was $\underline{\quad}\%$ at $\underline{\quad}\%$ coverage. Figure 2 will show the complete coverage-risk curve.

7.4 Ablation Results

Independent frame noise produced a UFFR of $\underline{\quad}\%$, compared with $\underline{\quad}\%$ under temporally correlated error replay. Removing confidence gating changed UFFR by $\underline{\quad}$ percentage points. The body-centred representation changed joint-angle MAE by $\underline{\quad}^\circ$. [Discuss only ablations whose confidence intervals and effect sizes support an interpretable difference.](#)

7.5 Fitness for Purpose

Of $\underline{\quad}$ evaluated model-exercise-outcome combinations, $\underline{\quad}$ met every registered fitness-for-purpose criterion, $\underline{\quad}$ were usable only with abstention, $\underline{\quad}$ failed the measurement criterion, and $\underline{\quad}$ remained unresolved because no defensible clinical

anchor was available. Replace this paragraph with the generated fitness-for-purpose summary and discuss its confidence intervals.

8 Discussion

This study asks a narrower and more consequential question than whether an egocentric pose looks plausible: whether the reconstructed signal supports a stable rehabilitation measurement and a defensible correction. The opening paragraph will state the answer using the registered criteria and the observed confidence intervals.

8.1 Geometric Accuracy Is Not Decision Reliability

The analysis is designed to reveal when reconstruction ranking and application ranking diverge. MPJPE weights every available joint equally and averages over time. A rehabilitation decision often depends on one angle at one phase of a repetition. The model with the best aggregate MPJPE may therefore produce a less reliable knee-depth decision if its error is concentrated at the knee during terminal flexion. Conversely, a model with moderate global error may preserve an angle because neighboring joints move coherently. The ground-truth and reconstructed branches in Fig. 1 make this difference measurable rather than anecdotal.

Use the observed model rankings to identify concrete rank reversals. Do not claim that MPJPE is uninformative if rankings agree for most outcomes.

8.2 Reliability Is Exercise and Outcome Specific

Clinical markerless-motion evidence does not support a single universal accuracy claim [11,13]. Our protocol therefore produces a matrix of model, exercise, outcome, and operating condition. A configuration may be adequate for repetition counting, which primarily depends on a low-frequency temporal pattern, while being inadequate for frontal-plane alignment feedback. This is not an inconsistency. It reflects different error tolerances and joint dependencies.

The same distinction should guide product behavior. If shoulder elevation remains stable but wrist orientation does not, the system can provide range-of-motion progress without issuing hand-placement corrections. Reliability-aware systems should expose this granularity instead of reducing the complete session to an undifferentiated “tracking quality” indicator.

8.3 Abstention as a Rehabilitation Function

Abstention is often treated as a lost prediction. In autonomous feedback, it is a safety function. A false correction can undermine trust and can prompt a user to alter an otherwise acceptable movement. The useful operating point is consequently not the threshold with maximum classification accuracy. It is the

threshold that limits unsupported feedback while retaining enough coverage to make the system useful.

Interpret the observed coverage-risk curve. State which visual or motion conditions trigger abstention, and distinguish calibrated uncertainty from simple joint-visibility heuristics.

If low UFFR is achieved only at negligible coverage, the method is not ready for autonomous feedback. It may still support asynchronous clinician review by selecting high-confidence summaries and attaching evidence frames. This graded interpretation is preferable to a binary claim that egocentric reconstruction either “works” or “does not work” for rehabilitation.

8.4 Implications for Feedback Generation

Separating structured findings from language generation creates an auditable boundary. Pose uncertainty affects whether a finding is supported; a language model affects how that finding is expressed. Combining both operations in a single unconstrained prompt makes it difficult to locate the cause of an erroneous instruction. It also permits a fluent model to present uncertain kinematics with unwarranted confidence.

Our primary deterministic realization avoids that failure mode. If LLM paraphrasing improves perceived clarity without changing anatomical content, it can be added as a presentation layer. It should not determine the body side, movement phase, or biomechanical error directly from a noisy coordinate stream. The dataset study can measure fidelity to the structured record, but a claim about therapeutic usefulness would require evaluation with patients and physiotherapists.

8.5 What the Cross-Dataset Result Establishes

The error-replay experiment establishes sensitivity under an empirically observed egocentric error process. It is stronger than adding arbitrary independent noise because it retains temporal correlation, missingness, speed dependence, and body-region structure. It is also transparent about its limit: the rehabilitation subjects were not recorded through the same head-mounted camera while performing the exercises.

A future paired dataset should combine head-mounted video, external video, laboratory motion capture, and independent physiotherapist ratings across repeated home-like sessions. Such a dataset would test visual domain shift, headset placement variation, and patient-specific movement directly. Until it exists, the present protocol offers a reproducible reliability envelope rather than a clinical deployment guarantee.

9 Limitations

First, REHAB24-6 and UCOPhyRehab contain external-camera recordings. Empirical error replay connects genuine egocentric reconstruction errors to their

reference motion, but it cannot reproduce correlations between a particular rehabilitation movement, head motion, home background, headset placement, and image quality. The study therefore does not claim direct egocentric video accuracy on these datasets.

Second, the public rehabilitation cohorts are small and consist primarily of healthy adults performing correct or deliberately incorrect movements. Their errors may not represent compensations, restricted range, tremor, fatigue, assistive devices, or movement strategies observed in patient populations. Technical agreement in these data is not evidence of clinical efficacy.

Third, OptiTrack and marker-derived joint centres are reference measurements rather than error-free anatomical truth. Soft-tissue artifact, marker placement, skeleton definitions, and synchronization contribute uncertainty. UCOPhyRehab instruments only selected joints for each exercise, limiting full-body conclusions.

Fourth, error replay assumes that residual windows can be transported between motions after body-frame alignment. Stratification by visibility, joint, and motion speed reduces this mismatch but cannot remove it. Results from synthetic egocentric data may also underestimate real-world domain shift; real-world residuals are therefore the primary replay source.

Fifth, the registered feedback rules operationalize a finite set of observable movement features. They do not diagnose injury, infer pain, determine an individualized exercise prescription, or replace professional judgment. Even a geometrically correct feedback message may be inappropriate for a particular patient.

Finally, statistical reliability is population and protocol dependent. ICC can change with between-subject heterogeneity, and MDC depends on the measurement distribution. We report absolute agreement, SEM, MDC95, and participant-level confidence intervals alongside ICC to avoid interpreting a single coefficient as universal reliability.

10 Conclusion

We introduced EgoRehab-Bench, a protocol for evaluating whether pose reconstructed from head-mounted video is sufficiently reliable for home rehabilitation assessment and feedback. The protocol separates direct egocentric reconstruction from rehabilitation decision sensitivity, replays temporally structured empirical errors on REHAB24-6 and UCOPhyRehab reference motion, and evaluates the complete path from joints to measurements, findings, and feedback. Its fitness-for-purpose rule requires acceptable measurement error, minimal detectable change, false-feedback risk, and retained coverage rather than relying on MPJPE alone.

The final conclusion will name the configurations that met these criteria and those that did not. The broader implication is independent of the numerical outcome: an egocentric reconstruction system should issue exercise feedback only for the body regions, movements, and conditions in which its uncertainty has been linked to downstream decision reliability. When that evidence is absent, abstention is a more trustworthy output than a plausible but unsupported correction.

Declarations

Ethics. This study uses existing datasets under their respective research licenses and does not involve new participant recruitment in its current form. **[TODO: Confirm whether institutional review or exemption documentation is required for secondary analysis.]**

Data availability. REHAB24-6 is publicly archived on Zenodo. UCOPhyRehab is available from its authors for research purposes. UnrealEgo2, UnrealEgoRW, and REST-HANDS are available under their respective research terms. Raw data will not be redistributed by the authors of this study.

Code availability. The adapters, registered configuration, analysis scripts, and generated tables will be released at an archival repository URL after anonymous review.

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Use of generative AI. Generative AI was used to assist with manuscript planning and language drafting. All study decisions, source verification, code, analyses, and conclusions remain the responsibility of the authors.

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